Atmo-Ensemble
Ensemble ML Research Proposal for Air Quality Multi-Classification

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Abstract

The objective of this initiative is to drive a research project in order to study ensemble models and compare the best implementation ways in a multi-class problem.

When selecting the theme of study, we took into consideration the constraints stated in the project description, which mentioned that we must select imbalanced datasets in a multi-class problem with at least 4 classes in the predicted feature, staying in the tabular side of machine learning techniques. Therefore, we decided to study air quality and pollution.

This dataset focuses on air quality assessment across various regions. The dataset contains 5000 samples and captures critical environmental and demographic factors that influence pollution levels, combining concentration of principal components that influence quality of air, weather conditions, and proximity to areas of risk.

One of our reasons to select this dataset was the fact that the labeled multi-class feature does not contain nominal information, but ordinal. The quality feature contains a value going from *Good* to *Hazardous*, divided into 4 different classes. These classes are connected between them as they have an order, and are not as if we classified animals or any other nominal class problem. This feature allows us to treat the data in different ways and get the best results.

Even if many interesting multi-class datasets are available on the internet fit for multi-class classification, most of them are too balanced, the objective feature contains 3 classes or is directly binomial classification, or is directly artificial. That is why after considering many options we decided to go with this idea.

1 Research Proposals

The main purpose of this project is to research; therefore, the objective is not just to get the best scores, but to understand what we are doing and why.

1.1 PCA and Dataset Feature Extraction

Does orthogonalizing correlated environmental features through Principal Component Analysis (PCA) and other feature extraction techniques improve the robustness and efficiency of ensemble classifiers, or does the transformation obscure the non-linear feature interactions that tree-based models naturally exploit?

Environmental datasets inherently suffer from multicollinearity. In this dataset, variables like coarse and fine particulate matter (PM_{10} and $PM_{2.5}$) are likely to be highly correlated, just as specific industrial gases (NO_2, SO_2, CO) might strongly co-occur based on the "Proximity to Industrial Areas" feature.

While tree-based ensembles are generally robust to multicollinearity when making predictions, highly correlated features can severely distort Feature Importance metrics (which directly affects Research Proposal 1.4) and add unnecessary computational overhead (which affects Research Proposal 1.6).

Furthermore, transforming the feature space X into principal components Z via the eigenvectors of the covariance matrix might smooth out noise, but it fundamentally changes how decision tree splits are calculated. Therefore, the objective is to evaluate how mathematical dimensionality reduction compares to using raw, interpretable sensor data within different ensemble architectures.

1.2 Benchmark Performance

Comparison of Bagging and Boosting

Does the variance-reduction of parallel ensembles offer better generalization on noisy sensor data compared to the bias-reduction of sequential ensembles?

We will compare the two fundamental techniques of ensemble learning: reducing variance (parallel) and reducing bias (sequential).

- **Parallel (Bagging)**: Train models independently to reduce overfitting (e.g., Random Forest, Extra Trees). Good for noisy sensor data.
- **Sequential (Boosting)**: Train models iteratively to correct previous errors (e.g., XGBoost, LightGBM, CatBoost). Good for finding difficult patterns.

Therefore, the objective is to compare Random Forest (Parallel) vs. Gradient Boosting (Sequential), introducing noise to the dataset (simulating sensor error) and seeing which architecture degrades faster.

1.3 Handling Ordinality

Does encoding ordinality via Regressor-based Ensembles outperform standard Multiclass Classifiers in Air Quality Assessment?

The objective feature is not a nominal class; if it contains an order, we aim to investigate if treating the target as ordinal ($Good < Moderate < Poor < Hazardous$) within an ensemble framework improves performance over treating it as purely nominal categories.

Our target variable isn't just A, B, C, D; it has a rank/order. Standard multiclass classification ignores this order (it treats "Good predicted as Hazardous" as an error equal to "Good predicted as Moderate"). Therefore, the objective is trying to modify ensemble objectives to respect order.

The approaches consist of:

- **Standard Multiclass**: Treat classes as nominal (independent).

- **Ordinal Ensembles**:
  - *Frank and Hall Method*: Decompose the 4 classes into 3 binary classifiers (Good vs. Rest, Moderate vs. Rest, etc.) and stack them.
  - *Regressor Ensemble*: Train a Regressor (XGBoost Regressor) to predict a continuous pollution score, then map the output to the closest class.

1.4 Feature Importance and Interactions (XAI)

Nowadays the understanding of the reasons for a model to make a decision are very relevant. We aim to use different techniques to interpret the models' decisions. We can elevate this by looking specifically at SHAP dependence plots to find non-linear interactions. For example, does high humidity compound the danger of high $PM_{2.5}$? Tree-based ensembles naturally capture these interactions, and SHAP can reveal if the models actually learned the chemical realities of smog formation.

1.5 Imbalanced Learning

1.5.1 Handling Rare Events

"Evaluating the efficacy of Hybrid Sampling-Ensemble methods (SMOTE-Boost vs. RUS-Boost) in detecting rare Hazardous Air Quality events."

Air quality data is naturally imbalanced. Hazardous samples are rare, but they are the most important to predict, as this is a recall-score critical problem. Standard accuracy is a bad metric here (a model that always predicts *Good* might be 0.9 accurate but useless in the real world).

The main idea is to integrate sampling techniques inside the ensemble algorithms. Techniques to compare:

- **Standard Ensembles**: Random Forest, AdaBoost.
- **Imbalanced-Specialized Ensembles**:
  - *Balanced Random Forest*: Down-samples the majority class in every tree.
  - *SMOTE-Boost*: Creates synthetic minority samples (SMOTE) before each boosting iteration.
  - *RUSBoost*: Uses Random Under Sampling (RUS) during boosting.

1.5.2 Cost-Sensitive Learning vs. Resampling Methods

A parallel approach to data resampling is algorithmic modification via Cost-Sensitive Learning. Instead of changing the data, we could change the loss function of the algorithm to heavily penalize dangerous mistakes. Because our target is ordinal, predicting "Hazardous" as "Good" is objectively worse than predicting "Hazardous" as "Poor". We will implement a custom loss function using a penalty matrix.

1.5.3 Robustness to Missing Data (Simulating Broken Sensors)

Air quality sensors fail constantly in the real world. Apart from adding noise, we will simulate data dropout. We will randomly drop 0.1, 0.2, or 0.3 of the $PM_{2.5}$ or NO_2 readings and compare two approaches: using a standard ensemble combined with an imputation technique (like k-NN or mean imputation) versus using XGBoost or LightGBM, which have native, built-in algorithms for handling NaN values by learning default split directions.

1.6 Efficiency vs. Accuracy

1.6.1 Trade-offs between Computational Cost and Classification Performance

Can Lightweight Ensembles replace Deep Stacking for Edge-based Air Quality Monitoring?

Air quality monitoring is often done on edge devices (small, low-power sensors in the city). Running a massive Stacking ensemble might be too heavy for a battery-powered chip. We aim to find the frontier between model size and performance.

- **Heavy Model**: A 3-layer Stacking Classifier (XGBoost + RF + SVM).
- **Light Model**: A highly pruned Decision Tree or a LightGBM with limited depth.
- **Measurement**: Measure not just Accuracy/F1-Score, but also Inference Time (ms) and Model Size (MB).

1.6.2 Conformal Prediction

Instead of forcing the model to output a single prediction (e.g., "Poor"), conformal prediction allows the model to output a set of predictions with a mathematical guarantee of containing the true class (e.g., predicting $\{\textit{Moderate, Poor}\}$ with 0.95 confidence).

We will study ways of changing the size of the prediction set between an uncalibrated Random Forest and a well-calibrated LightGBM and how it reacts when faced with noisy sensor data.

$$L(y, \hat{y}) = \sum_{i=1}^{N} C(y_i, \hat{y}_i)\,(y_i - \hat{y}_i)^2$$

Therefore, we look into the drawbacks of modifying the loss function in order to heavily penalize hazardous quality miss classifications and understanding if ordinal costs outperform synthetic data generation.

2 Dataset Feature Analysis

The features we are going to work with are the following:

- **Temperature** (°C): Average temperature of the region.
- **Humidity** (%): Relative humidity recorded in the region.
- $PM_{2.5}$ **Concentration** ($\mu g/m^3$): Fine particulate matter levels.
- PM_{10} **Concentration** ($\mu g/m^3$): Coarse particulate matter levels.
- NO_2 **Concentration** (ppb): Nitrogen dioxide levels.
- SO_2 **Concentration** (ppb): Sulfur dioxide levels.
- CO **Concentration** (ppm): Carbon monoxide levels.
- **Proximity to Industrial Areas** (km): Distance to the nearest industrial zone.
- **Population Density** (people/km^2): Number of people per square kilometer in the region.

Target Variable: Air Quality Levels

- **Good**: Clean air with low pollution levels.
- **Moderate**: Acceptable air quality but with some pollutants present.
- **Poor**: Noticeable pollution that may cause health issues for sensitive groups.
- **Hazardous**: Highly polluted air posing serious health risks to the population.

3 State of the Art and Previous Approaches

Most community members used tabular machine learning models individually, including Logistic Regression, Decision Trees, Naive Bayes, Support Vector Machines (SVM), GBM, Extreme Gradient Boosting (XGBoost), and MLP, achieving best results with Decision Trees (0.95+ F1 score). Even if Decision Trees represent a consensus model of many decision trees, we aim to approach the problem using ensemble models in different ways, comparing techniques to gain insights into our research objectives.